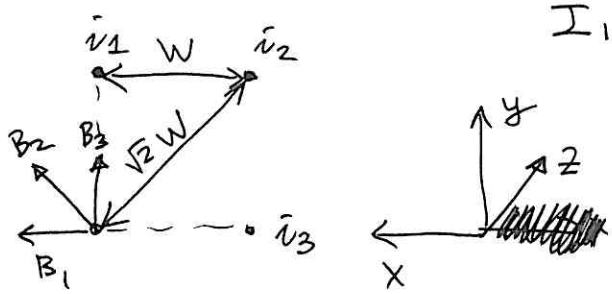
(1)

$$\vec{I}_{1,2,3} = i_{1,2,3}\,\hat{z}$$



(a)

$$\vec{B} = \frac{\mu_0 I}{4\pi}\int \frac{d\vec{l}\,' \times (\hat{r} - \hat{r}\,')}{|\vec{r} - \vec{r}\,'|^2}$$

$$\vec{B}_1 = \frac{\mu_0 i_1}{2\pi W}(+\hat{x}) \qquad \vec{B}_3 = \frac{\mu_0 i_3}{2\pi W}(\hat{y})$$

$$\vec{B}_2 = \frac{\mu_0 i_2}{2\pi(\sqrt{2}W)}\left(+\frac{1}{\sqrt{2}}\hat{x} + \frac{1}{\sqrt{2}}\hat{y}\right) = \frac{\mu_0 i_2}{8\pi W}(+\hat{x} + \hat{y})$$

$$\vec{B} = \frac{\mu_0}{2\pi W}\left\{+\left(i_1 + \tfrac{1}{2}i_2\right)\hat{x} + \left(\tfrac{1}{2}i_2 + i_3\right)\hat{y}\right\}.$$

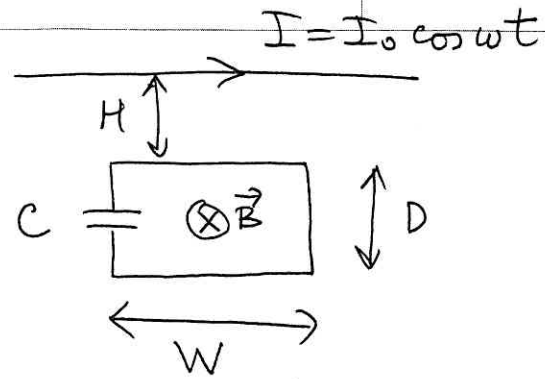
(b) Force on current $\vec{i}_4 = i_4(-\hat{z})$.

per unit length

$$\vec{F}_4 = \vec{i}_4 \times \vec{B} = \frac{\mu_0}{2\pi W}\left\{-\left(i_1 + \tfrac{1}{2}i_2\right)\hat{y} + \left(i_3 + \tfrac{1}{2}i_2\right)\hat{x}\right\}.$$

$$\hat{z}\times\hat{x} = \hat{y}\ ,\ \hat{z}\times\hat{y} = -\hat{x}$$

(2)



(a) $\mathcal{E} = -\dfrac{d\phi}{dt}$

$\phi = \int \vec{B} \cdot d\vec{A}$, $\vec{B}$ is into the page.

$B = \dfrac{\mu_0 I}{2\pi s}$, $dA = W\,ds$

$$\Rightarrow \phi = \int_H^{H+D} \frac{\mu_0 I}{2\pi s} W\,ds = \frac{\mu_0 W}{2\pi} I \ln\left(\frac{H+D}{H}\right)$$

where $I = I_0 \cos\omega t$.

$$\Rightarrow \mathcal{E} = -\frac{d\phi}{dt} = \frac{\mu_0 W}{2\pi} \ln\left(\frac{H+D}{H}\right) \omega I_0 \sin(\omega t).$$

(b) $Q = CV = C\mathcal{E}$

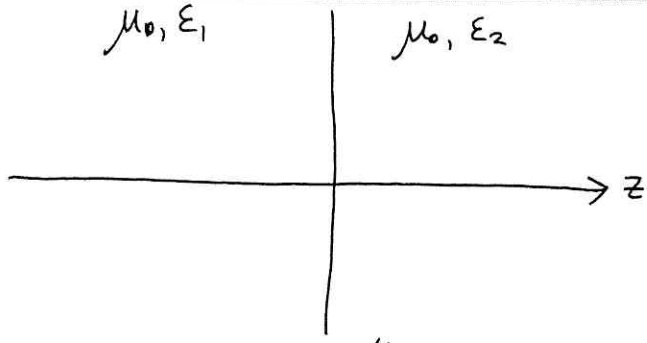
$$Q(t) = \frac{\mu_0 W}{2\pi} \ln\left(\frac{H+D}{H}\right) C \omega I_0 \sin(\omega t).$$

(c) $\omega \ll \dfrac{c}{D+W}$ means that frequency is much slower than inverse time during which EM wave propagates over the circuit.

Hence EM wave wavelength $\lambda = \dfrac{2\pi c}{\omega} \gg \dfrac{2\pi c}{\frac{c}{D+W}} = 2\pi(D+W)$

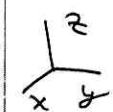
which means that the field over the circuit is uniform and displacement current $\dfrac{\partial \vec{E}}{\partial t}$ can be neglected $\Rightarrow$ $\vec{B}$ is determined by current I only.

(3)



$$\left.\begin{aligned} \vec{B}_i(z,t) &= \hat{y}\, B_{0i}\, e^{i(k_1 z-\omega t)} \\ \vec{B}_R(z,t) &= \hat{y}\, B_{0R}\, e^{-i(k_1 z+\omega t)} \end{aligned}\right\} z<0$$

$$\vec{B}_t(z,t) = \hat{y}\, B_{0t}\, e^{i(k_2 z-\omega t)} \quad \text{for } z>0$$



(a) $\vec{\nabla}\times\vec{B} = \mu\varepsilon \dfrac{\partial \vec{E}}{\partial t}$ (for $\vec{J}_f = 0$)

$$\Rightarrow i\vec{k}\times\vec{B} = \mu\varepsilon(-i\omega)\vec{E} \Rightarrow \vec{E} = -\frac{\vec{k}\times\vec{B}}{\omega\mu\varepsilon}$$

From wave equation $\nabla^2\vec{B} = \mu\varepsilon\dfrac{\partial^2\vec{B}}{\partial t} \Rightarrow V = \dfrac{1}{\sqrt{\mu\varepsilon}} = \dfrac{1}{\sqrt{\mu_0\varepsilon_0}}\sqrt{\dfrac{\mu_0\varepsilon_0}{\mu\varepsilon}} = \dfrac{c}{n}$

$$n = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}} = \sqrt{\varepsilon_{r1}} \text{ here.}$$

Thus, $\vec{E}_i = -\dfrac{k_1\hat{z}\times\hat{y}}{\frac{\omega}{V_1^2}} B_{0i}\, e^{i(k_1 z-\omega t)} = \dfrac{V_1^2 k_1}{\omega} B_{0i}\,\hat{x}\, e^{i(k_1 z-\omega t)}$

$\hat{z}\times\hat{y} = -\hat{x}$

$$= V_1 B_{0i}\,\hat{x}\, e^{i(k_1 z-\omega t)}$$

since $V_1 k_1 = \omega$.

and $V_1 = \dfrac{c}{n_1} = \dfrac{c}{\sqrt{\varepsilon_{r1}}}$.

$$\vec{E}_R = -\frac{k_1(-\hat{z})\times\hat{y}}{\frac{\omega}{V_1^2}} B_{0R}\, e^{-i(k_1 z+\omega t)} = V_1 B_{0R}(-\hat{x})\, e^{-i(k_1 z+\omega t)}$$

$$\vec{E}_t = -\frac{k_2\hat{z}\times\hat{y}}{\frac{\omega}{V_2^2}} B_{0t}\, e^{-i(k_2 z-\omega t)} = V_2 B_{0t}\,\hat{x}\, e^{i(k_2 z-\omega t)}$$

(b) Boundary conditions (1) $B_1^{\perp} = B_2^{\perp}$ → but here $B^{\perp} = B_z = 0$ on both sides.

(2) $\frac{1}{\mu_1} B_1^{\parallel} = \frac{1}{\mu_2} B_2^{\parallel}$ but here $\mu_1 = \mu_2 = \mu_0$

$\Rightarrow B_1^{\parallel} = B_2^{\parallel}$ at $z=0$. Parallel to the interface component is y here.

$$B_{0i} + B_{0R} = B_{0t} \quad (1')$$

(3) $\varepsilon_1 E_1^{\perp} = \varepsilon_2 E_2^{\perp}$ but here $\underset{\underset{E_z}{\parallel}}{E^{\perp}} = 0$ on both sides.

(4) $E_1^{\parallel} = E_2^{\parallel} \Rightarrow V_1 (B_{0i} - B_{0R}) = V_2 B_{0t}$ (2')

From (1') and (2')

$$\begin{cases} B_{0i} + B_{0R} = B_{0t} \\ B_{0i} - B_{0R} = \frac{V_2}{V_1} B_{0t} \end{cases} \Rightarrow \begin{cases} 2 B_{0i} = \left(1 + \frac{V_2}{V_1}\right) B_{0t} \\ 2 B_{0R} = \left(1 - \frac{V_2}{V_1}\right) B_{0t} \end{cases}$$

$$\frac{B_{0t}}{B_{0i}} = \frac{2V_1}{V_1+V_2} = \frac{2\frac{c}{n_1}}{\frac{c}{n_1} + \frac{c}{n_2}} = \frac{2}{n_1} \frac{n_1 n_2}{n_1+n_2} = \frac{2n_2}{n_1+n_2}$$

$$\frac{B_{0R}}{B_{0t}} = \frac{V_1 - V_2}{2V_1} \Rightarrow \frac{B_{0R}}{B_{0i}} = \frac{B_{0R}}{B_{0t}} \frac{B_{0t}}{B_{0i}} =$$

$$= \frac{V_1 - V_2}{2V_1} \frac{2V_1}{V_1+V_2} = \frac{V_1 - V_2}{V_1+V_2} =$$

$$= \frac{\frac{c}{n_1} - \frac{c}{n_2}}{\frac{c}{n_1} + \frac{c}{n_2}} = \frac{n_2 - n_1}{n_2 + n_1}$$

Thus $\frac{B_{0R}}{B_{0i}} = \frac{\sqrt{\varepsilon_{r2}} - \sqrt{\varepsilon_{r1}}}{\sqrt{\varepsilon_{r2}} + \sqrt{\varepsilon_{r1}}}$; $\frac{B_{0t}}{B_{0i}} = \frac{2\sqrt{\varepsilon_{r2}}}{\sqrt{\varepsilon_{r2}} + \sqrt{\varepsilon_{r1}}}$.

(c) $\vec{E}_i = \hat{x}\, E_{0i}\, e^{i(k_1 z - \omega t)}$ with $E_{0i} = v_1 B_{0i}$

$\vec{E}_R = +\hat{x}\, E_{0R}\, e^{-i(k_1 z + \omega t)}$ with $E_{0R} = -v_1 B_{0R}$

$\vec{E}_t = \hat{x}\, E_{0t}\, e^{i(k_2 z - \omega t)}$ with $E_{0t} = v_2 B_{0t}$

$$\Rightarrow \quad \frac{B_{0R}}{B_{0i}} = -\frac{E_{0R}/v_1}{E_{0i}/v_1} = -\frac{v_1}{v_1}\,\frac{E_{0R}}{E_{0t}}$$

$$\Rightarrow \quad \frac{E_{0R}}{E_{0i}} = -\frac{B_{0R}}{B_{0i}} = -\frac{n_2 - n_1}{n_2 + n_1} = \frac{n_1 - n_2}{n_1 + n_2} =$$

$$= \frac{\sqrt{\varepsilon_{r1}} - \sqrt{\varepsilon_{r2}}}{\sqrt{\varepsilon_{r1}} + \sqrt{\varepsilon_{r2}}}$$

Next, $\dfrac{B_{0t}}{B_{0i}} = \dfrac{E_{0t}/v_2}{E_{0i}/v_1} = \dfrac{v_1}{v_2}\,\dfrac{E_{0t}}{E_{0i}}$

$$\Rightarrow \quad \frac{E_{0t}}{E_{0i}} = \frac{v_2}{v_1}\,\frac{B_{0t}}{B_{0i}} = \frac{n_1}{n_2}\,\frac{2\,n_2}{n_1 + n_2} = \frac{2\,n_1}{n_1 + n_2} =$$

$$= \frac{2\sqrt{\varepsilon_{r1}}}{\sqrt{\varepsilon_{r1}} + \sqrt{\varepsilon_{r2}}}.$$

(d) Now $n_2 > n_1$, $\sqrt{\varepsilon_{r2}} > \sqrt{\varepsilon_{r1}}$.

$$\Rightarrow \quad \frac{B_{0R}}{B_{0i}} > 0 \quad \text{but} \quad \frac{E_{0R}}{E_{0i}} < 0.$$

Upon reflection, E_{0R} changes sign but also $\vec{k}_1$ changes sign $\Rightarrow$ B_{0R} has the same sign as incoming wave, B_{0i}.